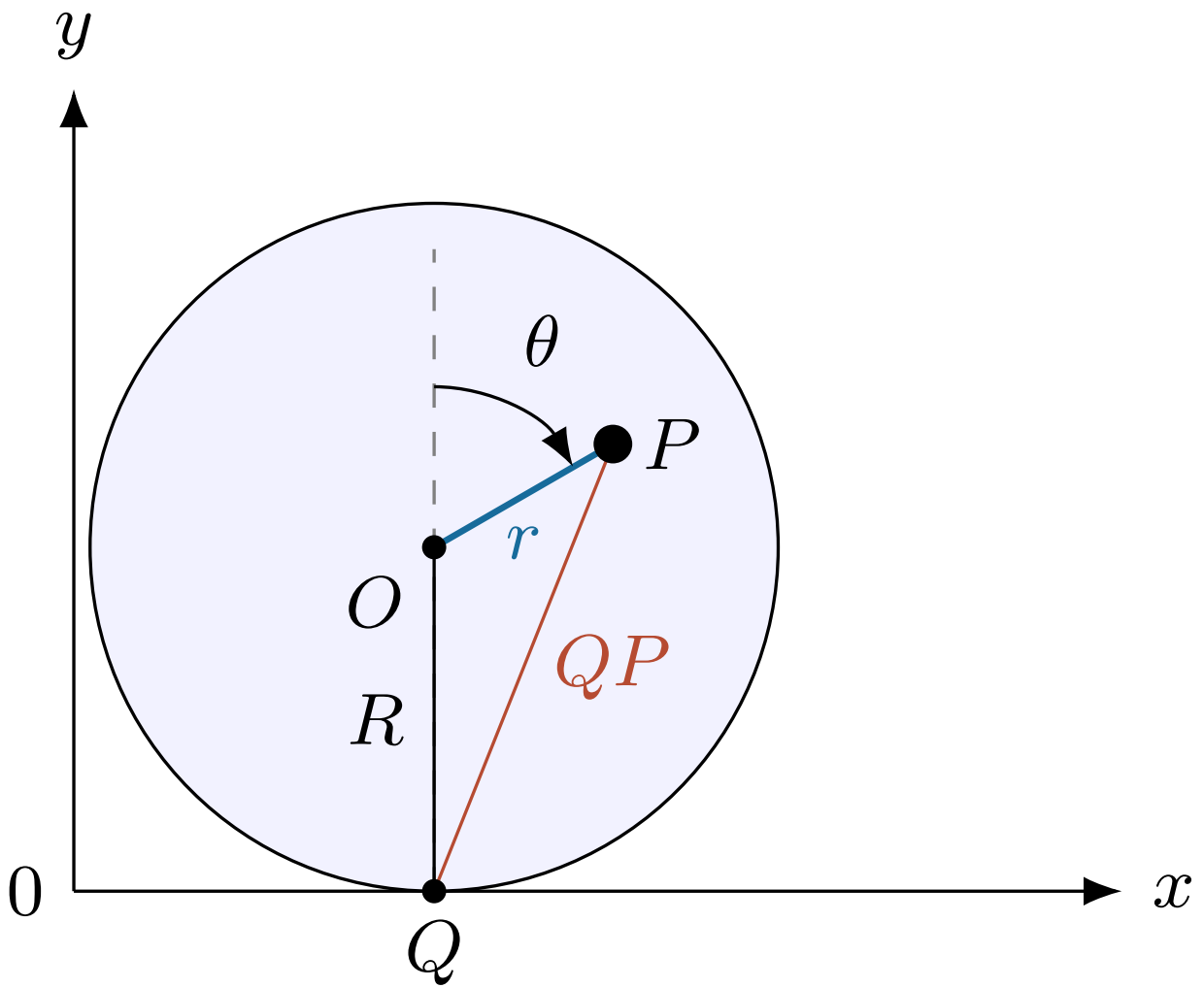

$$x_O = R\theta, \quad y_O = R$$